# NON-NEWTONIAN CALCULUS APPLIED TO PROBABILITY, UTILITY, AND BAYESIAN ANALYSIS

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## INTRODUCTION

A substantial body of evidence indicates that people really don't process probability information in the way that the classical or Bayesian models suggest. Some of this evidence is reviewed by Edwards (1954), Tversky and Kahneman (1974), and Slovic and Lichtenstein (1971). The purpose of this paper is to present a new theory of probability that is adapted to human behavior and decision making. Two basic ideas will be used: that probabilities do not obey the laws of ordinary arithmetic and calculus, but instead are governed by the laws of one of the Non-Newtonian Calculi and its corresponding arithmetic; and that if the entropies of two continuous probability distributions are equal, then those distributions (if unimodal) have psychologically equivalent degrees of dispersion. With these two ideas it is possible to reform the calculus of probabilities without abandoning the basic formalism of probability; only the numerical consequences of the traditional operations will be changed.

## NON-NEWTONIAN ARITHMETIC AND CALCULUS

Suppose that we have a collection of variables and quantities that do not obey the laws of ordinary arithmetic. E.g., it may not be permissible to "add" two similar quantities by classically adding their numerical values. An example of this is in the theory of Special Relativity, where we do not simply add the velocity-of-B-relative-to-A ($V_{BA}$) to the velocity-of-C-relative-to-B ($V_{CB}$) in order to obtain the velocity-of-C-relative-to-A ($V_{CA}$); it is not true that $V_{CA}=V_{CB}+V_{BA}$; instead we must use the colinear velocity composition rule $V_{CA}=(V_{CB}+V_{BA})/(1 + (V_{CB}V_{BA}/c^2))$, where c = the speed of light. It turns out that this rule can be written in the form:

$$V_{CA}/c = \tanh[\tanh^{-1}(V_{CB}/c) + \tanh^{-1}(V_{BA}/c)], \qquad (1)$$

which is equivalent to:

$$\tanh^{-1}(V_{CA}/c) = \tanh^{-1}(V_{CB}/c) + \tanh^{-1}(V_{BA}/c) \qquad (2)$$

If we are prepared to think of $\tanh^{-1}(V_{CA}/c)$ as a sort of generalized velocity-of-C-relative-to-A, then Eq. (2) really amounts to a simple addition rule for generalized velocities. But if we do not wish to alter our concept of velocity, then perhaps we can reinterpret Eq. (1) as a non-linear addition rule for velocities; i.e., if we define

$$x \dot{+} y \equiv \tanh(\tanh^{-1}x + \tanh^{-1}y), \qquad (3)$$

then (1) becomes simply

$$(V_{CA}/c) = (V_{CB}/c) \dot{+} (V_{BA}/c) . \qquad (4)$$

Eq. (3) is the definition of the addition operation in what we might call "tanh-arithmetic". Similarly we could define subtraction, division, and multiplication operations in tanh-arithmetic; and after that, a corresponding derivative (the limit of the tanh-ratio of the tanh-differences) and an integral (as the inverse of the tanh-derivative). But there's no reason to confine ourselves to tanh-arithmetic and tanh-calculus; let's generalize to β-arithmetic and β-calculus.

The basic idea of non-Newtonian arithmetic and calculus is to first *transform* the variables in such a way that they obey the rules of ordinary classical arithmetic and calculus. Then use ordinary arithmetic and calculus on the transformed variables, doing the sort of operation (addition, subtraction, division, multiplication, differentiation, or integration) that you originally wanted to do to the untransformed variables. Finally, take the result of this operation and transform it back (the inverse way) to get the solution. Thus, if $w \equiv \beta(z)$, where β is a monotonic function of z, with the unique inverse representation $z \equiv \beta^{-1}(w)$, then we define [see *Non-Newtonian Calculus* (1972), Chapter 5] the operations of β-arithmetic as follows:

$$\begin{aligned} x \dot{+} y &\equiv \beta[\beta^{-1}(x) + \beta^{-1}(y)] , \\ x \dot{-} y &\equiv \beta[\beta^{-1}(x) - \beta^{-1}(y)] , \\ x \dot{\times} y &\equiv \beta[\beta^{-1}(x) \times \beta^{-1}(y)] , \\ x \dot{/} y &\equiv \beta[\beta^{-1}(x) / \beta^{-1}(y)] . \end{aligned} \qquad (5)$$

It is a worthwhile exercise for the reader to confirm that the operations $\dot{+}$, $\dot{-}$, $\dot{\times}$, and $\dot{/}$ manipulate in exactly the same formal way as the classical arithmetic operators +, −, ×, and /; hence, if our intuition with respect to a system of variables says that arithmetic operations and manipulations such as we are familiar with in ordinary arithmetic must be possible, then we can conclude that our intuition does not, to that extent at least, rule out the possibility of arithmetic within our system being performed according to (5), for some β.

Suppose now that we want to form a derivative of x with respect to t, where x obeys the laws of β-arithmetic and t obeys the laws of α-arithmetic. We first apply $\beta^{-1}$ to x and $\alpha^{-1}$ to t, to get transformed variables $\beta^{-1}(x)$ and $\alpha^{-1}(t)$--both of which obey the laws of classical arithmetic--and then form the ordinary derivative of $\beta^{-1}(x)$ with respect to $\alpha^{-1}(t)$ to get $d\beta^{-1}(x)/d\alpha^{-1}(t)$. Finally, apply β to this derivative to get the non-Newtonian derivative of x with respect to t. (We do not apply α to $d\beta^{-1}(x)/d\alpha^{-1}(t)$; if we did so, we would not obtain a derivative with the proper--i.e., desired--manipulative properties.) This non-Newtonian deri-

vative is denoted $D^*x(t)$:

$$D^*x(t) \equiv \beta[d\beta^{-1}(x)/d\alpha^{-1}(t)] \ . \qquad (6)$$

By an analogous line of reasoning [for full details, see Grossman and Katz (1972), Chapter 6] we develop the non-Newtonian definite integral of x with respect to t, and arrive at the expression:

$$\int_{t_1}^{*t_2} x \qquad (7)$$

$$\equiv \beta\left\{\int_{\alpha^{-1}(t_1)}^{\alpha^{-1}(t_2)} \beta^{-1}[x(\alpha[z])]d\alpha(z)\right\}.$$

The non-Newtonian derivative and integral thus derived have the same manipulative properties as the Newtonian derivative and integral, though of course they have different numerical values, in general.

ENTROPY AND EQUIVALENT UNIFORM WIDTH

If $\tilde{x}$ is a random variable with classically defined probability density function f(x), then the Entropy of $\tilde{x}$, S(f), is defined:

$$S(f) \equiv -\int_{-\infty}^{+\infty} f(x) \log_e f(x)\, dx \ , \qquad (8)$$

where I have arbitrarily decided to use natural logs. The fact that S(f) is defined as a classical integral is of no particular importance; what matters is that S(f) measure a certain characteristic of f; it doesn't matter what mode of calculation is used to generate S(f) so long as S(f) has the appropriate properties.

If f(x) is any uniform distribution of width W, then

$$S(f) = \log W. \qquad (9)$$

If f(x) is a normal distribution with standard deviation $\sigma$, then

$$S(f) = \log[\sigma\sqrt{2\pi e}\,]. \qquad (10)$$

There is good reason for regarding S(f) as a uniquely desirable measure of "spread" or "uncertainty" of a distribution function [see Shannon and Weaver (1949), Khinchin (1957), Lees (1966), and Jaynes (1968) for some of the reasons.]. Suppose we inquire as to which uniform distribution has the same entropy as the normal distribution with standard deviation $\sigma$. Equating (9) and (10) immediately gives:

$$W = \sigma\sqrt{(2\pi e)} \cong (4.132731354)\sigma \qquad (11)$$

or, dividing (11) by 2 to get the half-width of the uniform:

$$(W/2) \cong (2.066365677)\sigma \ ; \qquad (12)$$

this is very nearly equal to the traditional "$\pm$ two standard deviations" of tolerance on either side of the mean of a normal distribution. In effect, the uniform distribution with the same entropy as a normal, assuming that both have the same mean, will "cut" the normal distribution at about the 95% confidence band width (the actual percentage is 96.120617%). This interpretation of the 95% confidence level (very popular indeed, when working with normal distributions), as merely being the *equivalent uniform width* of the normal (using entropy as the measure of equivalence), provides the key which, with the non-Newtonian derivative, will enable us to determine the appropriate arithmetic and calculus for probabilities.

A NON-NEWTONIAN CALCULUS FOR PROBABILITIES

I assume that β-arithmetic is the proper arithmetic for manipulating probabilities. I assume that all the customary rules of probability calculations are valid, except that they are restated in terms of β-arithmetic instead of classical arithmetic. Thus the relation between the joint probability P(y, x), the conditional P(y|x), and the prior P(x) is:

$$P(y,x) = P(y|x) \dot{\times} P(x) \qquad (13)$$
$$\equiv \beta[\beta^{-1}(P(y|x)) \times \beta^{-1}(P(x))];$$

and the rule for computing a marginal probability is [here the random variable $\tilde{y}$ can assume any of the values $y_1, y_2, \ldots, y_m$ and the random variable $\tilde{x}$ can assume any of the values $x_1, x_2, \ldots, x_n$]:

$$P(y_j)$$
$$= P(y_j,x_1) \dot{+} P(y_j,x_2) \dot{+} \ldots \dot{+} P(y_j,x_n)$$
$$= \beta[\beta^{-1}P(y_j,x_1)+\beta^{-1}P(y_j,x_2)+\ldots$$
$$+\beta^{-1}P(y_j,x_n)] \ , \ j=1,\ldots,m. \qquad (14)$$

Substitute (13) into (14), and simplify to get:

$$P(y_j) = \beta[\sum_{i=1}^{n} \beta^{-1}(P(y_j\ x_k))$$
$$\times \beta^{-1}(P(x_i))], \quad j=1,\ldots,m. \qquad (14)'$$

Bayes' Rule is:

$$P(x_i|y_j) = P(y_j|x_i) \dot{\times} P(x_i) \dot{/} P(y_j)$$
$$= \beta\left[\frac{\beta^{-1}(P(y_j|x_i))\times\beta^{-1}(P(x_i))}{\beta^{-1}(P(y_j))}\right],$$
$$i=1,\ldots,n, \quad j=1,\ldots,m; \qquad (15)$$

substitute (14)' into (15) to obtain:

$$P(x_i|y_j)$$
$$= \beta\left\{\frac{[\beta^{-1}(P(y_j|x_i))\times\beta^{-1}(P(x_i))]}{\sum_{k=1}^{n}[\beta^{-1}(P(y_j|x_k))\times\beta^{-1}(P(x_k))]}\right\},$$
$$i=1,\ldots,n, \ j=1,\ldots,m. \qquad (16)$$

This is the general form of Bayes' Rule. In (14), let $y_1 \equiv$ Success, and let $y_2 \equiv$ Failure, for m=2. Define:

(Utility of $x_i$) $\equiv U(x_i)$

$$\equiv P(\text{Success}|x_i); \qquad (17)$$

(Utility of a gamble in which one receives $x_i$ with probability $P(x_i)$, $i=1,\ldots,n$)

$$\equiv U(\text{Gamble}) \qquad (18)$$
$$\equiv P(\text{Success}|\text{Gamble}) \ .$$

But

$$P(\text{Success}|\text{Gamble}) = P(\text{Success}) \qquad (19)$$
$$= (\text{marginal probability of success}) \ .$$

Substitute (18) into (19), and substitute this with (17) into (14) to get (after applying $\beta^{-1}$ to both sides of (14)), for $j = 1$:

$$\beta^{-1}(U(\text{Gamble})) = \sum_{i=1}^{n} \beta^{-1}(U(x_i)) \times \beta^{-1}(P(x_i)). \qquad (20)$$

But any monotonic increasing function of a utility function, U, is also a utility function (so long as the monotonicity is strict throughout the relevant range of U) [see Chapter I of Meginniss (December 1977)]; so define:

$$V(x_i) \equiv \beta^{-1}(U(x_i)), \ i=1,\ldots,n \qquad (21)$$

and similarly for U(Gamble):

$$V(\text{Gamble}) \equiv \beta^{-1}(U(\text{Gamble})) \qquad (22)$$

Now substitute (22) and (21) into (20) to obtain:

$$V(\text{Gamble}) = \sum_{i=1}^{n} V(x_i) \times \beta^{-1}(P(x_i)), \qquad (23)$$

which is nothing less than the "Subjective Expected Utility Rule" [see discussion by Ward Edwards (1954) ].

All of the above results depend upon knowing what the function $\beta$ is. To determine $\beta$ we assume two things:

1. The most commonly encountered distribution "in nature" is the normal distribution (due to the Central Limit Theorem). Therefore the human brain should be "hard-wired" to process the normal distribution efficiently. Suppose that an individual encounters a Cumulative Normal distribution function in some real situation; the individual will probably wish to compute the density distribution corresponding to the encountered distribution; the density is determined by taking the non-Newtonian derivative of the cumulative distribution (since the cumulative values are actually probabilities, we know that $\beta$-calculus applies to the cumulative function). Let the cumulative be expressed as

$$F_N(t|\mu,\sigma) = (1/(\sqrt{2\pi}\sigma)) \int_{-\infty}^{t} \exp[(\hat{t}-\mu)^2/(2\sigma^2)]d\hat{t} \ . \qquad (24)$$

The most desirable density to compute from this integral would be a *uniform* density! I.e., we should pick $\beta$ so that the non-Newtonian derivative of $F_N(t|\mu,\sigma)$ with respect to t is a uniform density (i.e., a constant, on its non-zero width). We assume that the variable t, with respect to which differentiation of $F_N$ is made, obeys ordinary laws of arithmetic and calculus, so that $\alpha$ is the identify function (see Eq. (6)).

2. Our second assumption is that the uniform density obtained by differentiating $F_N$ should have the same mean, $\mu$, as $F_N$ and should have the same entropy--i.e., the width W of the uniform should satisfy (12).

These two conditions, plus the constraints

$$\beta(0) = 0$$
$$\beta(1) = 1 \qquad (25)$$
$$\beta(.5) = .5,$$

are sufficient to determine the function $\beta$ and its inverse, $\beta^{-1}$, if we assume that the non-Newtonian integral of the area beneath the uniform density is equal to 1.

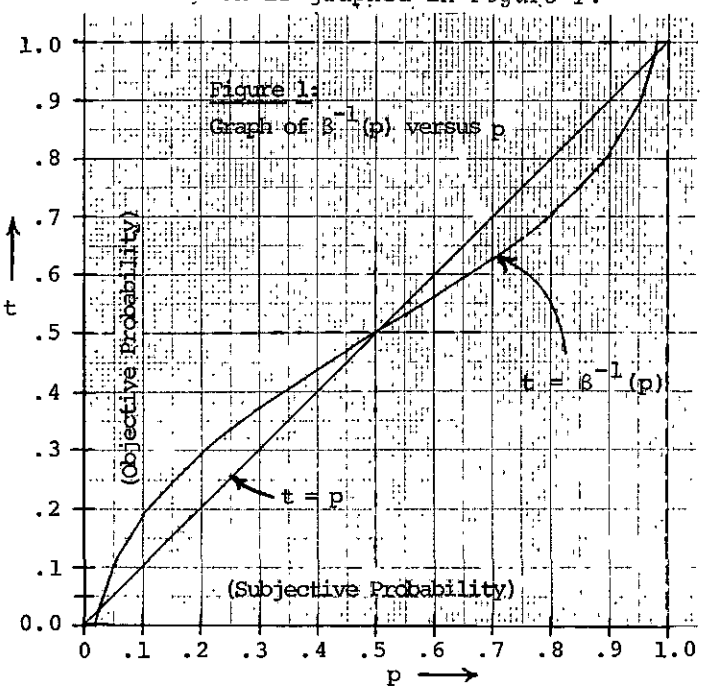
The solution is:

$$\beta(t) = \begin{cases} 0, \text{ if } t=0; \\ F_N(\sqrt{(2\pi e)}\ (t-.5) \ | \ 0, 1), \\ \quad \text{if } 0 < t < 1; \\ 1, \text{ if } t = 1. \end{cases} \qquad (26)$$

The inverse of (26) is:

$$\beta^{-1}(p) = \begin{cases} 0, \text{ if } 0 \le p \le F_n(-.5\sqrt{(2\pi e)}|0,1) \\ \qquad = .019396915; \\ [.5+\{F_N^{-1}(p|0,1)/(\sqrt{(2\pi e)}\}], \\ \quad \text{if } F_N(-.5\sqrt{(2\pi e)}|0,1) < p \\ \quad < F_N(+.5\sqrt{(2\pi e)}|0,1) \ ; \\ 1, \text{ if } F_N(+.5\sqrt{(2\pi e)}|0,1) \le p \le 1, \\ \quad \text{i.e., if } .980603085 \le p \le 1. \end{cases} \qquad (27)$$

This function is graphed in *Figure 1*.



This function is applied to calculating the sum of mutually exclusive event probabilities, and the product of independent event probabilities, in *Tables 1* and *2*, respectively.

Because the calculus of probabilities utilizing this function $\beta(t)$ and its inverse $\beta^{-1}(p)$ is fundamentally related to the Gaussian (or Normal) probability distribution, it might be appropriate to call this the Gauss Calculus and the various operations the Gaussian operations (e.g., Gaussian sum, Gaussian derivative).[2] This terminology would distinguish the present calculus from other conceivable non-Newtonian calculi.

SUBJECTIVE AND OBJECTIVE PROBABILITIES

If f is the "objective relative frequency" (or "objective probability") of an event, then the corresponding "subjective probability" of the event is equal to $\beta(f)$. To see this, consider an ordinary 6-sided fair die: each of the six distinct faces is equally likely to appear when the die is rolled; let $p_i$ = the subjective probability that face i will appear, i=1,2,...,6. Clearly $p \equiv p_1=p_2=\ldots=p_6$. Since subjective probabilities obey (by hypothesis) the Gaussian probability calculus, we have

$$\begin{aligned} 1 &= p_1 \dot{+} p_2 \dot{+} \ldots \dot{+} p_6 \\ &= \beta(\beta^{-1}(p_1) + \beta^{-1}(p_2) + \ldots + \beta^{-1}(p_6)) \\ &= \beta(6\cdot\beta^{-1}(p)) , \end{aligned}$$

therefore, applying $\beta^{-1}$ to both sides and using $\beta^{-1}(1) = 1$, we get

$$\begin{aligned} \beta^{-1}(p) &= 1/6, \quad \text{or} \\ p &= \beta(1/6) = \beta(.166667) \\ &= 0.084167. \end{aligned}$$

But p here is the subjective probability corresponding to the objective probability 1/6. We infer that the same general relationship obtains for other subjective and objective probabilities, i.e.,

$$\text{Subjective Probability} = \beta(\text{Objective Probability}) . \quad (28)$$

The supposition here is, of course, that Subjective Probabilities obey the Gaussian ($\beta$) calculus whereas Objective Probabilities obey classical calculus.

(28) justifies the labelling of the axes in *Figure 1*.

EXAMPLE: THE "CANCER TEST"

A familiar elementary example of the application of Bayes' Rule is the following: a screening test is performed on an asymptomatic patient in order to determine if the patient has a certain type of cancer. Let C denote the case where the Cancer is present in the patient and let H denote the case where the patient is Healthy (no cancer); the test result can be either positive (+), indicating that cancer is present, or negative (-), indicating that there is no cancer. Assume that the conditional probabilities are:

$$\begin{aligned} P(+|H) &= .04, \\ P(-|H) &= .96, \\ P(-|C) &= .05, \\ P(+|C) &= .95, \end{aligned} \quad (29)$$

and assume that the prior probabilities are:

$$\begin{aligned} P(C) &= .03 \\ P(H) &= .97. \end{aligned} \quad (30)$$

An individual takes the test and the test result is positive (+). What is the posterior probability that the individual has cancer? The traditional solution treats probabilities (29) and (30) as objective probabilities for which the traditional Bayes' Rule is applicable:

Table 1: The $\beta$-sum of the Probabilities of Two Mutually Exclusive Events: $p_1 \dot{+} p_2 \equiv \beta(\beta^{-1}(p_1)+\beta^{-1}(p_2))$ (Numerical entries for $p_1 \geq p_2$ only)

black box contains pairs $(p_1, p_2)$ for which $(p_1 \dot{+} p_2) \geq (p_1 + p_2)$ .

$(p_1 \dot{+} p_2)$ values are printed in *Italics*.

$(p_1 + p_2)$ values are printed in Regular Characters.

| $p_2$ ↑ / $p_1$ → | .01 | .02 | .05 | .1 | .2 | .3 | .4 | .5 | .6 | .7 | .8 | .9 | .95 | .98 | .99 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| .5 | | | | | | | | *1.00* / 1.00 | | | | | | | |
| .4 | | | | | | | *.941* / .8 | *.965* / .9 | *1.00* / 1.00 | | | | | | |
| .3 | | | | | | *.845* / .6 | *.901* / .7 | *.938* / .8 | *.964* / .9 | *1.00* / 1.00 | | | | | |
| .2 | | | | | *.648* / .4 | *.758* / .5 | *.834* / .6 | *.889* / .7 | *.930* / .8 | *.960* / .9 | *1.00* / 1.00 | | | | |
| .1 | | | | *.310* / .2 | *.477* / .3 | *.603* / .4 | *.703* / .5 | *.784* / .6 | *.850* / .7 | *.905* / .8 | *.948* / .9 | *1.00* / 1.00 | | | |
| .05 | | | *.111* / .10 | *.195* / .15 | *.337* / .25 | *.459* / .35 | *.567* / .45 | *.663* / .55 | *.750* / .65 | *.828* / .75 | *.897* / .85 | *.956* / .95 | *1.00* / 1.00 | | |
| .02 | | *.021* / .04 | *.051* / .07 | *.102* / .12 | *.203* / .22 | *.304* / .32 | *.405* / .42 | *.505* / .52 | *.604* / .62 | *.704* / .72 | *.804* / .82 | *.902* / .92 | *.951* / .97 | *1.00* / 1.00 | |
| .01 | *.00* / .02 | *.020* / .03 | *.050* / .06 | *.100* / .11 | *.200* / .21 | *.300* / .31 | *.400* / .41 | *.500* / .51 | *.600* / .61 | *.700* / .71 | *.800* / .81 | *.900* / .91 | *.950* / .96 | *.980* / .99 | *1.00* / 1.00 |

$$P(C|+) = \left\{\frac{(.95)(.03)}{(.95)(.03)+(.04)(.97)}\right\} = .4235; \qquad (31)$$

The answer, .4235, is also regarded as an objective probability. If, however, we assume that probabilities (29) and (30) are subjective probabilities (say, as assessed by the patient), then the Gaussian version of Bayes' Rule, (16), must be used:

$$\begin{aligned} P(C|+) &= \beta\{[\beta^{-1}(.95)\cdot\beta^{-1}(.03)]/ \\ &\quad [\beta^{-1}(.95)\cdot\beta^{-1}(.03) \\ &\quad + \beta^{-1}(.04)\cdot\beta^{-1}(.97)]\} \qquad (32) \\ &= \beta\left\{\frac{(.898)(.045)}{(.898)(.045)+(.076)(.955)}\right\} \\ &= \beta(.358) = .278; \end{aligned}$$

the answer, .278, is a subjective probability, and the corresponding objective probability is, by (28), $\beta^{-1}(.278)=.358$, which is to be contrasted with .4235 obtained in (31). We thus see that if probabilities (29) and (30) are mistakenly regarded as objective rather than subjective probabilities, then the Bayes' calculation will give misleading results. A third possibility is that the probabilities (29) are objective (e.g., they may be based upon empirical results in a large number of trial tests with medical follow-up to determine if cancer actually developed) whereas the probabilities (30) are subjective (merely representing the patient's initial introspective opinion); the Bayes' calculation should then be:

$$\begin{aligned} P(C|+) &= \beta\left\{\frac{(.95)\cdot\beta^{-1}(.03)}{(.95)\cdot\beta^{-1}(.03)+(.04)\beta^{-1}(.97)}\right\} \\ &= \beta\left\{\frac{(.95)\ (.045)}{(.95)\ (.045)\ +\ (.04)\ (.955)}\right\} \\ &= \beta(.5281) = .5462\ ; \qquad (33) \end{aligned}$$

this answer is obviously also a subjective probability, since $P(C|+)$ is only a subjective revision, in the light of the

Table 2:

The $\beta$-product of the Probabilities of Two Independent Events:

$p_1 \dot{\times} p_2 \equiv \beta[\beta^{-1}(p_1) \times \beta^{-1}(p_2)]$

(Numerical entries for $p_1 \geq p_2$ only).

($p_1 \dot{\times} p_2$) values are printed in *Italics*.

($p_1 \times p_2$) values are printed in Regular Characters.

The Annular area between the two black boxes contains the pairs ($p_1$, $p_2$) for which ($p_1 \dot{\times} p_2$)>($p_1 \times p_2$).

| $p_2$ \ $p_1$ | .01 | .02 | .05 | .1 | .2 | .3 | .4 | .5 | .6 | .7 | .8 | .9 | .95 | .98 | .99 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | *1.000* |
| .99 | | | | | | | | | | | | | | | .9801 |
| | | | | | | | | | | | | | | *.979* | *.980* |
| .98 | | | | | | | | | | | | | | .9604 | .9702 |
| | | | | | | | | | | | | | *.897* | *.949* | *.950* |
| .95 | | | | | | | | | | | | | .9025 | .931 | .9405 |
| | | | | | | | | | | | | *.741* | *.826* | *.898* | *.900* |
| .9 | | | | | | | | | | | | .81 | .855 | .882 | .891 |
| | | | | | | | | | | | *.493* | *.614* | *.708* | *.798* | *.800* |
| .8 | | | | | | | | | | | .64 | .72 | .76 | .784 | .792 |
| | | | | | | | | | | *.329* | *.404* | *.513* | *.603* | *.697* | *.700* |
| .7 | | | | | | | | | | .49 | .56 | .63 | .665 | .686 | .693 |
| | | | | | | | | | *.222* | *.270* | *.332* | *.425* | *.506* | *.597* | *.600* |
| .6 | | | | | | | | | .36 | .42 | .48 | .54 | .570 | .588 | .594 |
| | | | | | | | | *.151* | *.182* | *.220* | *.270* | *.347* | *.417* | *.498* | *.500* |
| .5 | | | | | | | | .25 | .30 | .35 | .40 | .45 | .475 | .490 | .495 |
| | | | | | | | *.102* | *.123* | *.147* | *.176* | *.215* | *.275* | *.331* | *.398* | *.400* |
| .4 | | | | | | | .16 | .20 | .24 | .28 | .32 | .36 | .38 | .392 | .396 |
| | | | | | | *.068* | *.082* | *.098* | *.115* | *.136* | *.163* | *.207* | *.248* | *.298* | *.300* |
| .3 | | | | | | .09 | .12 | .15 | .18 | .21 | .24 | .27 | .285 | .294 | .297 |
| | | | | | *.044* | *.054* | *.063* | *.073* | *.084* | *.097* | *.114* | *.141* | *.167* | *.199* | *.200* |
| .2 | | | | | .04 | .06 | .08 | .10 | .12 | .14 | .16 | .18 | .19 | .196 | .198 |
| | | | | *.028* | *.033* | *.038* | *.043* | *.047* | *.052* | *.058* | *.065* | *.076* | *.087* | *.097* | *.100* |
| .1 | | | | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 | .095 | .098 | .099 |
| | | | *.022* | *.024* | *.026* | *.028* | *.030* | *.032* | *.034* | *.036* | *.038* | *.042* | *.046* | *.050* | *.050* |
| .05 | | | .0025 | .005 | .010 | .015 | .020 | .025 | .030 | .035 | .040 | .045 | .0475 | .0049 | .0495 |
| | | *.019* | *.019* | *.020* | *.020* | *.020* | *.020* | *.020* | *.020* | *.020* | *.020* | *.020* | *.020* | *.020* | *.020* |
| .02 | | .0004 | .001 | .002 | .004 | .006 | .008 | .010 | .012 | .014 | .016 | .018 | .019 | .0196 | .0198 |
| | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* | *.000* |
| .01 | .0001 | .0002 | .0005 | .001 | .002 | .003 | .004 | .005 | .006 | .007 | .008 | .009 | .0095 | .0098 | .0099 |

test result, of the prior probability P(C); the objective probability corresponding to .5462 is $\beta^{-1}(.5462) = .5281$, which is to be compared with .4235 obtained in (31) and .358 obtained in (32). The importance of determining just what kind of probabilities are being used (i.e., objective or subjective) is apparent.

I should note that the calculation (33) bears a strong resemblance to a subjective version of Bayes' Rule incorporating a so-called "impediment function" to modify the values of some of the probabilities, as used by Frank Lad (1978, Eq. (1)); the basic difference being that Lad transforms the conditional rather than the prior probabilities before utilizing the traditional form of Bayes' Rule. (Also, Lad's approach is not imbedded in a general theory of probability.)

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FOOTNOTES

[1]About two years ago I discovered that if a velocity composition rule $V_{CA}=f(V_{CB}, V_{BA})$ is assumed to exist, and if $V_{RS}=-V_{SR}$, and if the Principle of Relativity is assumed to hold [so that on solving for $V_{CB}$ we obtain $V_{CB}=f(V_{CA}, V_{AB})$, where $f(\cdot\ ,\ \cdot)$ is the same function as before] then it follows that $f(V_{CB}, V_{BA}) \equiv g[g^{-1}(V_{CB}) + g^{-1}(V_{BA})]$. To obtain (1) it is necessary to make additional assumptions, such as that the speed of light is invariant. But <u>every</u> velocity composition rule consistent with the Principle of Relativity must be of the form given here, namely $V_{CA} = g[g^{-1}(V_{CB}) + g^{-1}(V_{BA})]$, where $g^{-1}$ is the unique inverse of g.

[2]This terminology was suggested to me by Michael Grossman and Robert Katz on May 3, 1980.

BIBLIOGRAPHY

Abramowitz, Milton, and Stegun, Irene. *Handbook of Mathematical Functions With Formulas, Graphs, and Mathematical Tables.* U.S. Department of Commerce, National Bureau of Standards. U.S. Government Printing Office, 1964. pp. 976-77. An indispensable reference.

Edwards, Ward. "The Theory of Decision Making." *Psychological Bulletin* 51 (1954): 4, pp. 380-417. Reprinted in Edwards, Ward, and Tversky, Amos (editors), *Decision Making*, Penquin Books, Ltd., Harmondsworth, Middlesex, England, 1967, pp. 13-64.

Grossman, Michael, and Katz, Robert. *Non-Newtonian Calculus.* Lee Press, Pigeon Cove, Mass. 01966. Especially chapters 5 and 6.

Jaynes, Edwin. "Prior Probabilities." *IEEE Transactions on Systems Science and Cybernetics*, Vol SSC-4, No. 3, Sept. 1968, pp. 227-41.

Khinchin, A.I. *Mathematical Foundations of Information Theory.* Dover Publications, N.Y. 1957. Contains a beautiful proof of the entropy--is-unique theorem.

Lad, Frank. "Embedding Bayes' Theorem in General Learning Rules: Connections Between Idealized Behaviour and Empirical Research on Learning." *British Journal of Mathematical and Statistical Psychology* 31 (1978), pp. 113-25.

Lees, Sidney, "Uncertainty and Imprecision." *Journal of Basic Engineering: Transactions of the ASME*, June 1966, pp. 369-78.

Meginniss, James R. *Alternatives to the Expected Utility Rule.* Unpublished doctoral dissertation. University of Chicago, Graduate School of Business. December 1977. Chapters I, II, and Appendix C.

_____. "A New Class of Symmetric Utility Rules for Gambles, Subjective Marginal Probability Functions, and A Generalized Bayes' Rule." *1976 Business and Economic Statistics Section Proceedings of the American Statistical Association.* pp. 471-6. An abbreviated version of Appendix C of Meginniss (1977).

_____. "Utility, Ignorance, and Impartiality: A Derivation of Diffuse Prior Probability Distributions.". *1978 Business and Economic Statistics Section Proceedings of the American Statistical Association.* pp. 757-62.

Shannon, Claude E., and Weaver, Warren. *The Mathematical Theory of Communication.* University of Illinois Press, Urbana, 1949. Reprinted as a University of Illinois paperback, 1969.

Slovic, Paul, and Lichtenstein, Sarah. "Comparison of Bayesian and Regression Approaches to the Study of Information Processing in Judgment." *Organizational Behavior and Human Performance* 6: 6 (Nov. 1971), p. 693.

Tversky, Amos, and Kahneman, Daniel. "Judgement Under Uncertainty: Heuristics and Biases." *Science* 185 (27 Sept. 1974), p. 1125.